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DCDC Power Conversion System in Electric Vehicle Part 2: Design and Development

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Executive Summary

The increasing demand to extend the driving range of electric vehicles—particularly in commercial sectors—has accelerated the development of high-voltage (HV) battery systems, with nominal voltages reaching up to 1000 V. Despite this advancement, numerous safety-critical and auxiliary subsystems, such as electric steering pumps, braking systems, and cabin heating units, continue to operate at low-voltage (LV) levels—typically 12 V for passenger vehicles and 24 V for commercial vehicles. Consequently, high-efficiency DC-DC converters are essential for reliable and effective power conversion from HV to LV domains. Moreover, the need to optimize the utilization of the vehicle's Energy Storage System (ESS), while maintaining compliance with rigorous Automotive Safety Integrity Level (ASIL) requirements, has intensified the demand for advanced DC-DC converter designs that exhibit high efficiency, enhanced power density, and improved thermal and electrical performance. Unlike industrial DC-DC converters, automotive-grade converters must meet significantly more stringent specifications, particularly with respect to mechanical vibration robustness, electromagnetic compatibility (EMC), and functional safety compliance, to ensure reliable and safe operation under demanding automotive environments.

***Keywords**:* battery-electric-vehicle (BEV), DC-DC converter, PSFB, DAB, voltage clamp

1 Introduction

The electrification of modern vehicles, particularly with the transition towards hybrid and fully electric powertrains, has driven an increased demand for highly efficient and reliable power conversion systems for converting the unregulated input high voltage (e.g., typically between 400V-1000VDC), provided from high voltage energy storage system (ESS) to a regulated output voltage typically 24V, with high conversion efficiency to supply low voltage loads in commercial vehicles. In automotive applications, high voltage DC-DC power converters align with low voltage batteries are fundamental components to provide low voltage power net [1]. Nevertheless, according to automotive standards, DC systems operating above 60VDC must incorporate functional (i.e., the output is isolated, but there is no protection against electric shock), basic (i.e., the isolation offers shock protection as long as the barrier is intact), or reinforced (i.e., a single barrier equivalent to two layers of basic insulation) isolation from lower voltage levels, whereas the degree of isolation dictated by the system high voltage requirements. Conversely, LV systems operating below 60VDC are typically grounded to the vehicle chassis to provide a stable electrical potential reference, which helps define voltage levels and maintain the integrity of the vehicle's electrical architecture [2]. Therefore, this requirement limits the automotive DC-DC converters to the isolated types. Figure 1 shows a typical battery electric vehicle (BEV) power train system and typical isolated DC-DC power converter. Further, the demand for low-voltage power supply is growing by emerging advanced safety-critical systems—such as steer-by-wire, brake-by-wire, and powerful electronic control units (ECUs) such as advanced driver assistance systems (ADAS), necessitates the development of DC-DC converters that offer enhanced power density and transient response while maintaining cost-efficiency and sustainability [2]. One effective approach to enhancing the power density of DC-DC converters, while simultaneously minimizing the size of passive components, including inductors, isolation transformers, and capacitors, is to increase the operating switching frequency. However, applying higher switching frequency enforced of employing advanced hardware (HW) and control solutions to keep up global efficiency of the DC-DC converter and preventing extra pressure on cooling system. Therefore, careful design consideration is needed to mitigate increased switching losses, the adverse effects of parasitic elements, and the occurrence of voltage overshoots and oscillations across power semiconductor devices. To address these challenges and sustain high overall efficiency, it is essential to employ advanced hardware solutions—such as wide-bandgap semiconductor devices—and to implement sophisticated soft-switching techniques, including Zero-Current Switching (ZCS) and Zero-Voltage Switching (ZVS) [3][4]. Nevertheless, the ability to simultaneously achieve high efficiency, reliability and reduced electromagnetic interference, and compact passive component sizing is limited to a select subset of converter topologies.

This paper presents a comprehensive review of the advantages and challenges associated with the most widely adopted high-frequency isolated DC-DC converter topologies in BEV applications. Section 1 provides an overview of the BEV powertrain architecture, emphasizing the role and functional requirements of DC-DC converters within this system. Section 2 offers a detailed examination of the predominant converter topologies employed in automotive applications, focusing on their structural and operational characteristics. Section 3 summarizes the key findings and insights derived from the review.

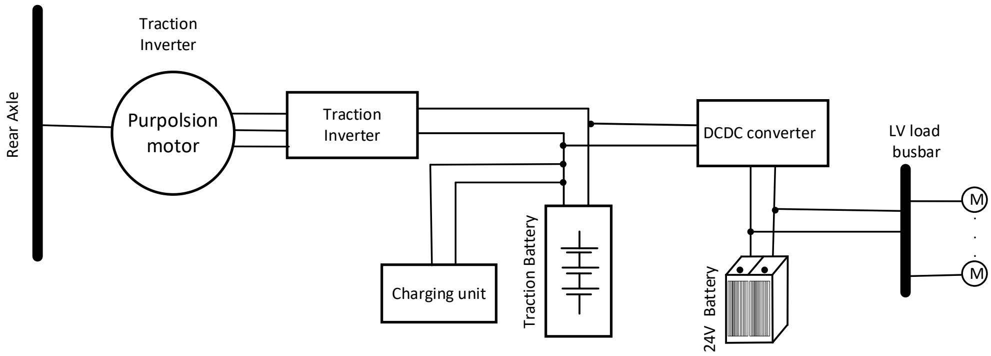


Figure 1. Typical commercial vehicle power train structure

2 DC-DC Converter Topologies for Automotive applications

In BEV applications, high-voltage DC-DC converters are employed to step down the traction-battery voltage to standardized low voltage levels, typically 12 V for passenger vehicles and 24 V for commercial vehicles. In accordance with established standards (e.g., STANDARD), these converters are also required to ensure adequate galvanic isolation between the high- and low-voltage domains. This isolation can be realized through either capacitive or inductive means. While capacitive isolation offers advantages in terms of power density and is generally suitable for low-power applications, inductive isolation, commonly implemented via high-frequency transformers, is preferred for high-power converters with substantial voltage conversion ratios due to its superior handling of power and voltage stress [5]. The three predominant topologies for transformer-based DC-DC converters are the Dual-Active Bridge (DAB), Phase-Shifted Full-Bridge (PSFB), and LLC Resonant Converter. These architectures are widely adopted in high-power applications due to their ability to provide galvanic isolation, support soft-switching techniques, and achieve high efficiency across varying load conditions [6]-[8]. Among these topologies, the LLC resonant converter uniquely enables soft switching on both the primary and secondary sides, thereby facilitating superior peak efficiency compared to the PSFB and DAB topologies. This efficiency advantage is primarily attributed to its ability to achieve ZVS and ZCS over a narrow operating range. However, this feature limits the performance of the LLC topology in applications requiring a wide input and output voltage range due to its strong dependence on the ratio between the transformer's magnetizing inductance and the resonant inductance. This sensitivity leads to diminished efficiency under off-nominal operating conditions and complicates regulation control across a broad voltage window. As a result, the LLC converter is not considered a viable solution for such use cases and thus is excluded from the scope of this review [9].

2.1 Dual-Active Bridge (DAB) Topology

Among the various characteristics of the Dual-Active Bridge (DAB) topology, the most prominent are its high-power handling capability, inherent bidirectional power transfer, and the facilitation of soft-switching operation [6][7]. As illustrated in Figure 2, the DAB DC-DC converter topology comprises two full-bridge configurations of power electronic switches, input and output DC-link capacitors, an auxiliary inductor, and a high-frequency isolation transformer. The transformer ensures galvanic isolation and enables voltage level adaptation between the high- and low-voltage sides, while its leakage inductance can function as the ZVS inductor to provide soft-switching transitions to reduce corresponding switching losses and EMI. In comparison to PSFB and LLC topologies, DAB uses fewer passive components which increases the overall power density.

The bidirectional operation of the DAB is enabled by introducing an additional phase shift between the primary and secondary full bridges. The sign of this phase shift determines the direction of power flow while the magnitude of the transferred power is dictated by the current flowing through the auxiliary inductor, as expressed by Equation (1) [8].

$$P = \frac{NV_1V_2\varphi(\pi-|\varphi|)}{2\pi^2F_SL} \tag{1}$$

The maximum theoretical phase angle between the phase voltages of the primary and secondary side in a DAB converter is 90°, at which point maximum power transfer occurs. However, a considerably increased efficiency can be achieved for a given DAB converter with the use of alternative modulation schemes. Figure 3(a) illustrates the relationship between the modulation scheme (selected phase-shift value) and the corresponding inductor size required on the primary side. Additionally, Figure 3(b) depicts the necessary DC blocking capacitor value as a function of the chosen inductor value, demonstrating the interplay between these two components in determining the system's performance [8].

As previously mentioned, the control algorithm in a Dual Active Bridge (DAB) converter utilizes two phase-shift angles. Angle 1: Intra-bridge angle, which refers to the phase shift between the half-bridges within the primary full-bridge. Angle 2: Inter-bridge angle, which refers to the phase shift between the primary and secondary full bridges [7]-[8].

Maximum power transfer occurs when Angle 1 and Angle 2 are set to 180° and 90°, respectively. However, in addition to phase-shift control, the DAB's power transfer capability is also influenced by the gain, which is a crucial parameter. The gain is typically defined as the ratio of the output voltage to the input voltage, and it governs how effectively the DAB converter transfers power. The gain value is given by [7][8]:

$$Gain = \frac{V_{or}}{V_{in}} \tag{2}$$

Where, V_{in} is an input volage, V_{or} is reflected output voltage to the primary side of the transformer. The DAB kernels are illustrated in Figure 4 (a). Although Figure 4(a) demonstrates that locking the gain to a higher value allows for greater power transfer to the secondary side, Figure 4(b) reveals that for certain ranges of Angle 1 and Angle 2, the RMS current on the primary side increases, leading to higher conduction losses. From a ZVS perspective, as depicted in Figure 4(c), the largest ZVS region for both the primary and secondary side MOSFET switches is achievable when Angle 1 is locked at 180°, and the gain is set to 1. This condition maximizes the ZVS region, thereby enhancing overall efficiency by reducing switching losses while maintaining optimal power transfer [7][8]. However, considering automotive system requirements, particularly regarding variable gain and load range, there is a necessary trade-off between the selected values of Angle 1, Angle 2, and the achievable ZVS. In general, keeping Angle 1 at 180° and maintaining a unitary gain provides the maximum ZVS. On the other hand, setting Angle 1 and Angle 2 to 90° at a gain of 1 offers the lowest RMS current. It is important to note that while increasing ZVS at turn-on can improve switching performance, it may also result in higher circulating RMS currents, leading to increased conduction losses. Therefore, an optimal balance between ZVS benefits and current management is crucial to meet the system's efficiency requirements [8].

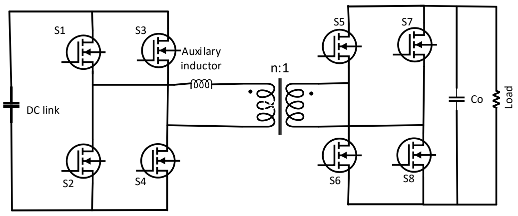
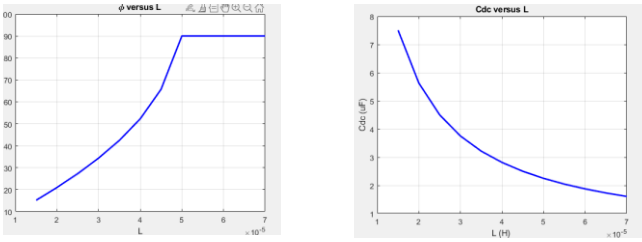
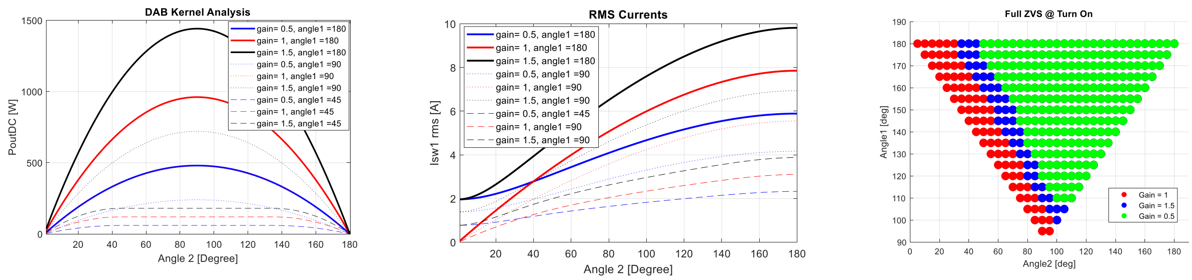


Figure 2. typical dual active bridge (DAB) DC-DC converter topology



(a) (b)

Figure 3. a) practical phase-shift value vs. auxiliary inductor value at the primary side, b) DC blocking capacitor value as a function of the chosen auxiliary inductor value.



(a) (b) (c)

Figure 4. a) The DAB kernel, the power curves. b) the RMS current curves at primary side, c) the full Zero Voltage Switching (ZVS) region [8].

2.2 Phase-shift full-bridge (PSFB) synchronous rectifier Topology

The Phase-Shifted Full-Bridge (PSFB) topology with synchronous rectification, illustrated in Figure 5, is widely adopted in automotive power conversion applications due to its favorable efficiency and isolation characteristics. Similar to the DAB topology, PSFB employs full-bridge configurations on both the primary and secondary sides of the isolation transformer. However, in contrast to DAB, output voltage regulation in PSFB is achieved through introducing only one phase angle between primary-side legs while the buck operation is facilitated by the output inductor (L_o) while resonance frequency of L_o is twice switching frequency. Furthermore, the PSFB topology is particularly well-suited for applications that require a higher current slew rate and benefit from the implementation of peak current mode control. Its inherent operational characteristics enable stable and precise current regulation, making it advantageous in scenarios where controlled current dynamics and robust protection against overcurrent conditions are critical [9]. To achieve high efficiency across a wide load range and accommodate the broad input and output voltage variations, particularly under high switching frequency operation aimed at enhancing power density, a meticulously engineered combination of hardware and software design is required. Such design must ensure the preservation of soft-switching conditions on the primary side to minimize switching losses, while simultaneously implementing effective strategies for suppressing voltage stress on the secondary-side rectifiers. This necessitates careful coordination of component sizing, control algorithms, and timing sequences to optimize overall converter performance and reliability [9]-[10].

2.2.1 ZVS at Primary-Side MOSFETs

The series inductor (L_s), in PSFB topology modulates the effective pulse width of the secondary-side voltage. Larger L_s results in increased duty cycle losses due to extended energy transfer intervals, thereby reducing the effective duty cycle (D_{eff}). To increase and enhance voltage regulation, a smaller L_s is preferred. However, a reduced inductance limits the energy storage capacity of the system, adversely affecting the soft-switching capability, particularly ZVS of the primary-side MOSFETs. Therefore, a critical design trade-off arises between maintaining robust soft-switching conditions and achieving a wide D_{eff} range for optimal converter performance [9][10]. Additionally, the output L_o significantly influences the soft-switching performance of the primary-side MOSFETs. A reduction in L_o leads to increased current ripple, which in turn diminishes the effectiveness of Zero- ZVS by reducing the available energy required to fully discharge the parasitic output capacitances of the MOSFETs during switching transitions. This degradation in soft-switching performance can result in elevated switching losses and reduced overall converter efficiency. The required primary-side current to achieve ZVS on the primary-side leg-1 MOSFETs ($I_{p\_leg1\_ZVS}$), as shown in (3), is primarily sustained by the reflected load current through the transformer, which is directly influenced by the peak-to-peak current ripple on L_o ($I_{Lo,pp}$), transformer turns ratio (n), and average output current ($I_{o,avg}$). Hence, careful optimization of L_o is essential to balance output voltage regulation, ripple current, and soft-switching performance [10].

$$I_{p\_leg1\_ZVS} = \frac{I_{o,avg} - \frac{I_{Lo,pp}}{2}}{n} \tag{3}$$

2.2.2 Reducing rectifier Voltage Stress

The series inductance, L_s resonates with the output capacitor (C_{oss}) of the secondary side full-bridge MOSFET, at given frequency by (4). results in larger rectifier voltage (V_{DS}) ringing while the peak value could be as high as ($\frac{2V_{in}}{n}$). The high dv/dt nature of this ringing voltage could be the origin of common mode noises and causing broadband EMI problems, the frequency of which is centered at the ringing frequency. Thereby achieving better efficiency and lowering the EMI noises, the rectifier voltage stress should be clamped. The primary side voltage clamp is provided by adding two diodes, with the L_s placed between the half-bridge FET leg and the clamping diodes. The clamping circuit schematic is shown in Figure 6(a) By doing so, the only series inductor, at primary-side, will be the transformer leakage inductor [9].

$$\omega = \frac{1}{\sqrt{2C_{oss}L_s}} \tag{4}$$

In addition, as illustrated in Figure 6(b), a passive resistor-capacitor diode clamp circuit can be integrated at the output stage to effectively mitigate voltage stress on the secondary-side rectifier MOSFETs. In this configuration, the clamping capacitor (C_{cl}) functions as a voltage source, which is charged through the clamping diode during voltage transients and subsequently dissipates the stored energy through the resistor (R_{cl}) [9]. The primary advantage of this approach lies in its simplicity and ability to suppress voltage spikes; however, it also presents notable design challenges. Specifically, to approximate ideal voltage source behavior, C_{cl} must be sufficiently large. This, in turn, necessitates a proportionally larger R_{cl} to ensure that energy dissipation occurs at the desired clamping voltage level (V_{CP}), which may limit the dynamic response and increase the passive component size—counteracting miniaturization goals in high-density power converter designs [9]. To overcome this challenge an active clamp circuit can be implemented in which the diode and passive resistor are replaced by the clamp MOSFET (Q_{cl}) which is inserted before the output inductor. The schematic is shown in Figure 6(c) The main advantage of this clamp circuit is that it adds complexity to the control algorithm and adds one more active switch to the circuit [9].

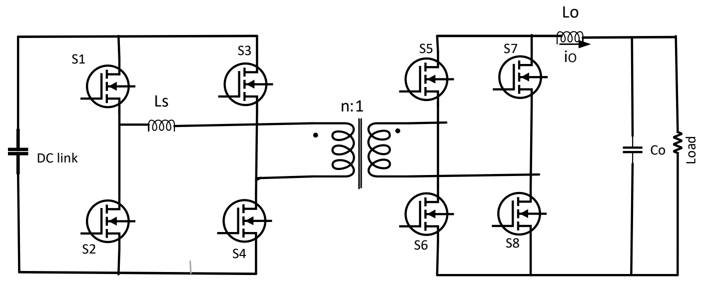
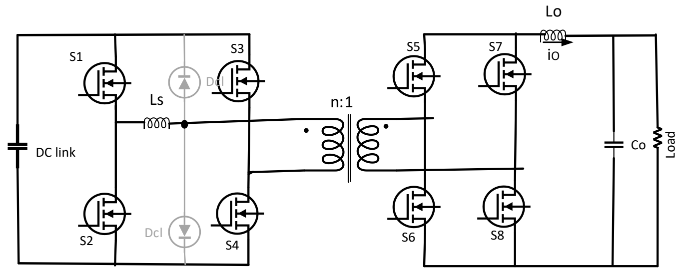
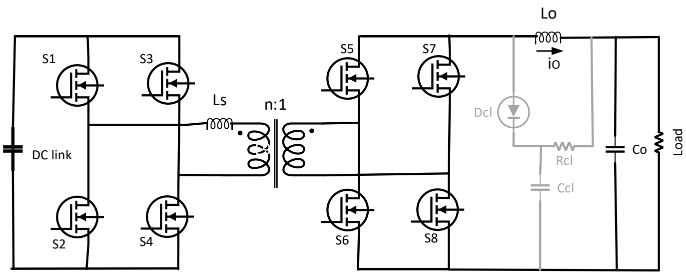


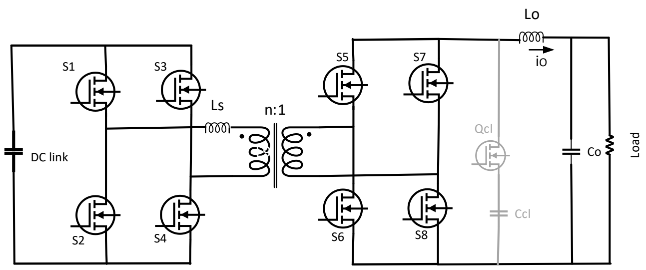
Figure 5. Typical topology of PSFB synchronous DC-DC converter



(a)



(b)



(c)

Figure 6. Typical voltage clamp circuits applied to PSFB topology, a) primary side diode voltage clamp; b) passive resistive-capacitor diode output voltage clamp; c) active output voltage clamp.

3 Conclusions

This paper provides a detailed review of the most widely adopted high-frequency isolated DC-DC converter topologies utilized in BEV applications. Among these, DAB and PSFB topologies are considered highly suitable for automotive environments, which are characterized by wide input and output voltage variations. Both topologies exhibit fast dynamic response to load transients, an essential feature for maintaining voltage stability under varying operational conditions.

The implementation of synchronous rectification on the secondary side in both DAB and PSFB architectures significantly reduces conduction losses, thereby enhancing overall efficiency. A distinct advantage of the DAB topology is its inherent capability for bidirectional power transfer at high power levels, making it particularly favorable for applications involving regenerative breaking or auxiliary power exchange.

Conversely, a notable limitation of the PSFB topology is the elevated voltage stress imposed on the secondary-side rectifier devices during switching transitions. This necessitates the incorporation of additional voltage clamp circuits to attenuate voltage overshoots and suppress electromagnetic interference (EMI), thereby ensuring reliable operation and compliance with automotive EMC standards.

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